Project Interim Report
On

Courier Delivery Tracking System
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming With Java

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Abstract

The *Courier Delivery Tracking System* is a Java-based application developed using core programming and object-oriented concepts. The main purpose of this project is to manage and track courier deliveries efficiently through a simple console-based system. It allows users to book a courier, generate a unique tracking ID, and monitor the delivery status in real time.
This project is needed to overcome the limitations of manual courier tracking systems, which are often slow, error-prone, and difficult to manage. By digitizing the process, it ensures better accuracy, faster access to information, and improved customer experience.
The system performs key operations such as adding new courier details, searching couriers using tracking IDs, updating delivery status, and displaying complete information. It uses classes like Courier, User, and CourierService to organize data and logic effectively. Overall, the project demonstrates how object-oriented programming can be used to build a practical and efficient delivery management system.

Introduction

The *Courier Delivery Tracking System* is a Java-based project developed using core Java and OOP concepts. It helps users to book and track courier deliveries easily. When a courier is booked, the system generates a unique tracking ID. Using this ID, users can check the delivery status anytime.
This project is useful because it reduces manual work and makes courier management faster and more accurate. It replaces traditional paper-based systems with a digital solution.
The system can perform operations like booking a courier, tracking its status, and updating delivery progress (Booked, In Transit, Delivered). It is designed using different classes to organize data and logic efficiently.

scope

This project is suitable for small courier services or basic tracking systems where parcel details can be managed easily using simple Java programs. It is designed for learning purposes and works well for small-scale applications without needing complex tools.

In the future, the project can be extended by adding database integration to store parcel data permanently. It can also be improved by adding features like user login, real-time tracking updates, graphical user interface (GUI), and online access. This will make the system more powerful and useful for larger courier companies.

System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum **4 GB RAM**
- Processor: Intel i3 or higher
- Storage: 100 MB free space

🔹 Software Requirements

- Java Development Kit **(JDK 8 or above)**
- IDE: Visual Studio Code / Eclipse IDE / EditPlus
- Operating System: Windows / Linux / macOS

Methodology

The project works in a step-by-step process:

First, we add parcel details by creating parcel objects with a unique parcel ID and an initial status like "Dispatched". These parcels are stored in a list.

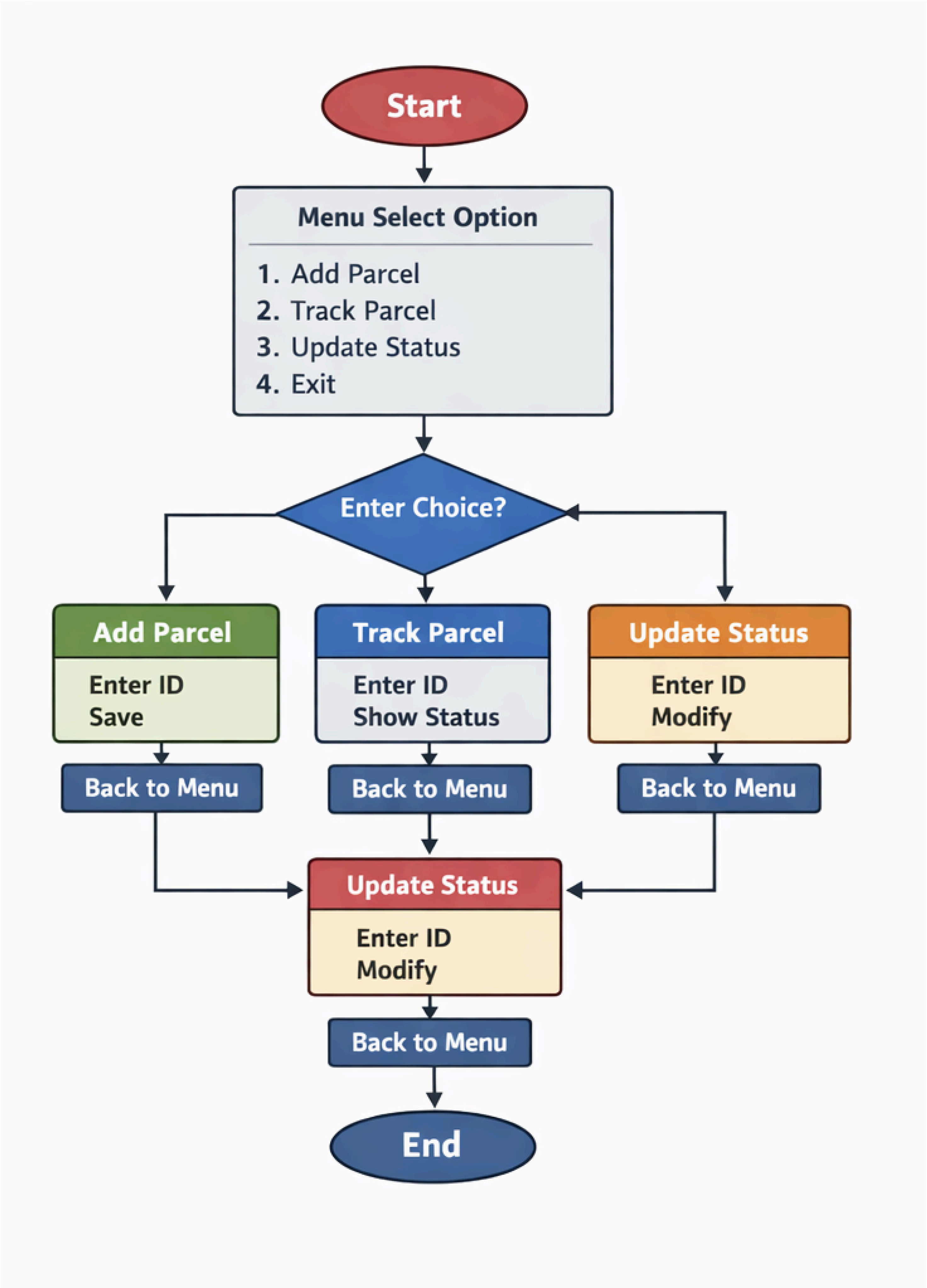
Next, the system allows the user to search for a parcel using its parcel ID. It checks all stored parcels and finds the matching one.

After finding the parcel, the system displays its current delivery status to the user.

Then, the user can modify or update the parcel status when needed, such as changing it to "In Transit", "Out for Delivery", or "Delivered".

Finally, the updated information is saved, and the user can repeat the process to track or manage other parcels easily.

Flowchart


Start
Menu Select Option
1. Add Parcel
2. Track Parcel
3. Update Status
4. Exit
Enter Choice?
Add Parcel
Enter ID
Save
Track Parcel
Enter ID
Show Status
Update Status
Enter ID
Modify
Back to Menu
Back to Menu
Back to Menu
Update Status
Enter ID
Modify
Back to Menu
End

Implementation

Classes Used (Logic Only)

- **Courier:** Stores courier details like tracking ID, sender, receiver, address, and status. Status is updated during delivery.
- **User:** Represents customer. Used to book and track couriers.
- **CourierService:** Core class that manages all operations such as adding courier, searching by tracking ID, and updating status.
- **TrackingSystem (Main):** Controls program flow, takes user input, displays menu, and calls service methods.
- **Employee (Optional):** Updates courier delivery status and handles deliveries.

```
Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLi
u want to re-enable it, run 'Import-Module PSReadLine'.

PS C:\Users\hp\OneDrive\Desktop\javabignnire> cd "c:\Users\hp\OneDrive\Desktop\javabignnire\"
 } ; if ($?) { java CourierSystem }
? Parcel Added: P101
? Parcel Added: P102
? Parcel Added: P103

Parcel List:
[P101] [P102] [P103]
[----] [----] [----]
[----] [----] [----]

Searching for Parcel ID: 2354
? Parcel Not Found

Enter Parcel ID to Update: 2494
? Parcel Not Found

Search Again (Enter ID): 387955
? Parcel Not Found
PS C:\Users\hp\OneDrive\Desktop\javabignnire> s
```

Future Enhancements

- Add database (MySQL)
- Create GUI (Swing/JavaFX)
- Add login system
- Enable real-time tracking
- Add SMS/Email notifications
- Integrate online payment

Gantt Chart

1. Planning & Structure

A Gantt chart helps in dividing the Courier Delivery Tracking System project into different tasks such as requirement analysis, class design, coding, and testing. It provides a clear structure of how the project is developed step by step. Even if the project is rebuilt or modified, proper planning is required.

2. Time Management

A Gantt chart shows when each task starts and when it finishes. This helps in managing time effectively and ensures that the project is completed on schedule. It also helps to avoid delays and last-minute panic.

Task	Duration
Requirement Analysis	1 day
Class Design	2 days
Coding (Add, Search, Update)	5 days
Testing & Debugging	3 days
Output & Screenshot	2 days
Report Preparation	2 days

Conclusion

The project successfully demonstrates the use of Java and Object-Oriented Programming concepts such as classes, objects, and methods. It provides a simple and functional system to track parcel delivery status. The project also helps in understanding real-world applications of programming and improves logical thinking and problem-solving skills.

References

References

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